Labsheet 2 : BTCS301P-AS02

Computer Vision Lab

Khushi Choudhary CU240251564

In [1]:
```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# 1. Load a low-contrast image in grayscale
# Replace 'low_contrast_image.jpg' with your image file path
img_gray = cv2.imread('low_contrast_image.jpg', cv2.IMREAD_GRAYSCALE)

# If no image file is present, create a synthetic low-contrast test image
if img_gray is None:
    img_gray = np.random.randint(80, 140, size=(300, 300), dtype=np.uint8)

# 2. Contrast Stretching (Min-Max Normalization to [0, 255])
min_val = np.min(img_gray)
max_val = np.max(img_gray)
if max_val > min_val:
    contrast_stretched = np.uint8(
        ((img_gray - min_val) / (max_val - min_val)) * 255
    )
else:
    contrast_stretched = img_gray.copy()

# 3. Global Histogram Equalization
hist_equalized = cv2.equalizeHist(img_gray)

# 4. Contrast Limited Adaptive Histogram Equalization (CLAHE)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
clahe_enhanced = clahe.apply(img_gray)

# 5. Visualization: Images and Histograms Comparison
images = [img_gray, contrast_stretched, hist_equalized, clahe_enhanced]
titles = [
    'Original (Low Contrast)',
    'Contrast Stretched',
    'Global Hist Equalization',
    'CLAHE',
]

plt.figure(figsize=(16, 8))

for i in range(4):
    # Plot processed images
    plt.subplot(2, 4, i + 1)
    plt.imshow(images[i], cmap='gray', vmin=0, vmax=255)
    plt.title(titles[i], fontsize=11, fontweight='bold')
    plt.axis('off')

    # Plot corresponding histograms
    plt.subplot(2, 4, i + 5)
    hist = cv2.calcHist([images[i]], [0], None, [256], [0, 256])
    plt.plot(hist, color='black', linewidth=1.2)
```

```
    plt.xlim([0, 256])
    plt.xlabel('Pixel Intensity')
    plt.ylabel('Frequency')
    plt.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```

